Gravitational Phenomena from Medium Flow: A Tick-Rate Framework with Atomic Mass Predictions

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Abstract

We present a framework that provides a physical cause for gravity: a medium flows toward mass, and atomic processes experience velocity-dependent resistance in that flow. Beginning from the empirical tick-rate gradient measured by atomic clocks at different gravitational potentials, we show that the unique flow field required to reproduce this gradient is $v(r) = \sqrt{2GM/r}$—the classical escape velocity. This result is not assumed but derived from matching clock data to a velocity-dependent mechanism. The combined effect of gravitational flow and kinematic motion defines an effective velocity $v_{\text{eff}} = \sqrt{v_{\text{esc}}^2 + v_{\text{motion}}^2}$ whose associated Lorentz factor controls all local tick rates. The framework is formally equivalent to General Relativity in the Painlevé–Gullstrand gauge: the Schwarzschild metric reduces to a single refractive index $n(r) = (1 - r_s/r)^{-1/2}$ controlling both temporal and spatial metric components. All 15 experimental tests of GR are reproduced with quantitative agreement.

The framework's natural variables are κ (the per-atom tick-rate contribution, measured from clocks) and N (the number of atoms). Newton's gravitational constant G is a translation constant between this language and the kilogram language of Newtonian gravity; it is not fundamental. We identify the quantum chromodynamic (QCD) vacuum as the candidate medium. Extending to sub-nuclear scales, we model baryon masses as medium processing costs (4 parameters, 12 baryons, $< 1.1\%$ RMS) and fit atomic masses from configuration alone (8 parameters, 43 atoms, 0.019% RMS). The proton mass emerges unconstrained at 938.272 MeV.

Nuclear binding energy per nucleon ($BE =$ 8.8 MeV for iron-56) is identified as the tick-rate suppression fraction $BE/m_p = 0.94\%$—numerically identical to the fractional mass defect. The returnion $r_{\text{ret}} = \kappa c^2 E_p/(BE\, c^2) = 1.324 \times 10^{-52}$ m marks the crossover scale between nuclear and gravitational dominance. Planetary mass follows from κ, N, and escape velocity alone, recovering Earth's mass to 0.0004%. The equivalence principle is structurally guaranteed: interaction and inertial mass share the same binding fraction, which cancels exactly.

The cosmological constant problem is dissolved: QFT vacuum energy calculations measure kinetic processing costs of matter fields, while Λ measures the medium's baseline potential tension—different physical quantities not subject to comparison. The framework predicts composition-dependent deviations from the equivalence principle at $\eta \sim 4 \times 10^{-18}$ for Ti vs. Pt, two orders below current sensitivity.

1 Introduction

1.1 On the authority of description

General Relativity is the most successful gravitational theory ever constructed. It passes every experimental test to which it has been subjected. None of what follows disputes this.

What follows disputes something else: the unearned step from descriptive success to ontological authority. General Relativity tells us *what* happens near mass with extraordinary precision. It does not tell us *why* or *how*. The field equations relate the stress-energy tensor to spacetime curvature, but the mechanism by which mass produces curvature is not derived—it is postulated. "Spacetime curves" is not an explanation. It is a mathematical description that has been granted the status of a physical claim without ever having to justify the promotion.

This matters because the authority of General Relativity does not end at its own predictions. It sets the terms by which new ideas are evaluated. Any alternative framework is immediately measured against GR's ontology—does it reproduce curved spacetime? Is it diffeomorphism invariant?—as though these are features of nature rather than features of one particular encoding of nature. The standard that new theories must satisfy is one that General Relativity itself never met: providing a physical mechanism for its central claim.

Physics was not always conducted this way. It began as physical interpretation—observing nature, proposing mechanisms, testing consequences. What started as the mathematical encoding of physical intuition has gradually inverted. Modern theoretical physics increasingly begins from mathematics and searches for physical content afterward. Dark energy is a case study: an unexpected observation (accelerating expansion) was absorbed into the existing framework by inventing a new entity to preserve that framework, rather than questioning whether the framework that made the observation unexpected might itself be incomplete. Billions in funding now flow toward characterizing an entity that may be an artifact of the model rather than a feature of nature.

The present paper takes the original approach. It starts from what is measured—atomic clock data, nuclear binding energies, hadronic masses—and asks what physical picture is consistent with those measurements. The result is a framework that reproduces every tested prediction of General Relativity through a different mechanism: a flowing medium rather than curved geometry. Its existence does not prove General Relativity wrong. It proves something more fundamental: that General Relativity's ontological claims are underdetermined by the evidence. Two different physical pictures produce the same predictions. The observations cannot distinguish between them. Neither picture is entitled to ontological authority on the basis of predictive success alone.

This is offered not as an attack on the standard framework but as a demonstration that the framework is one description among possible descriptions—a powerful and elegant one, but a description nonetheless. If the mere existence of a working alternative unsettles the field, that says more about the field's relationship with its assumptions than about the alternative.

1.2 The framework

General Relativity describes gravitational phenomena through the curvature of a four-dimensional spacetime manifold. The mathematical framework is experimentally confirmed to high precision, yet it offers no physical mechanism for why mass curves spacetime. The metric tensor encodes the geometry; the Einstein field equations relate it to the stress-energy tensor; but the causal chain from mass to curvature to motion remains postulated

rather than derived.

Gravity is a force. Objects fall. Clocks slow near mass. Orbits precess. These are observed facts, not in dispute. General Relativity describes all of them with extraordinary precision. This paper does not dispute GR's predictions.

This paper asks a question GR does not answer: *why does gravity exist?* GR postulates that mass curves spacetime. It does not explain why mass curves spacetime. The curvature is asserted, not caused. The field equations are a relationship, not a mechanism.

This paper proposes a mechanism. Space is a medium that flows toward mass. Atomic processes experience velocity-dependent resistance in this flow. The result is time dilation, spatial contraction, and gravitational acceleration—not as geometric postulates but as physical consequences of motion through a medium. The quantum chromodynamic (QCD) vacuum is identified as the candidate medium.

The mechanism's natural language is not Newton's gravitational constant G and bulk mass M. It is κ—the per-atom tick-rate contribution, measured directly from atomic clocks—and N, the number of atoms. Newton needed G because he could not see atoms; he had to package the collective gravitational behavior of $\sim 10^{50}$ atoms into a single constant. In this framework, G is a translation constant between the atom-counting language of the medium and the kilogram language of Newtonian gravity. It is not fundamental. The force it describes is real; the constant itself is a unit conversion.

The framework makes five claims, each with independent evidence:

1. **Formal equivalence with GR.** The Schwarzschild metric tensor is exactly a tick-rate field defined by one refractive index $n(r)$. Geodesics become Fermat's principle. All experimental GR tests are reproduced (Sec. 4).

2. **Atomic masses from configuration.** Modeling nucleon masses as quark processing costs and nuclear binding from the semi-empirical mass formula, 8 parameters fit 43 atomic masses to 0.019% RMS without using measured atomic masses as input. The proton mass emerges as an output (Sec. 5).

3. **Hierarchy resolution.** The strong force, electromagnetism, and gravity are three scales of the same medium processing, partitioned by local potential (Sec. 6).

4. **Cosmological constant dissolved.** The discrepancy between QFT vacuum energy and the observed Λ is not a fine-tuning problem but a category error: the two quantities measure kinetic processing costs and medium potential tension respectively—fundamentally different things (Sec. 9.7).

2 The Medium

2.1 Mass defect as tick-rate suppression

Nuclear binding energy has been precisely measured for decades. Gravitational time dilation has been measured independently with equal precision. The medium framework places these two results side by side and identifies them as the same phenomenon observed at different scales.

For iron-56, the binding energy per nucleon is $BE = 8.8$ MeV. The fractional tick-rate suppression this represents relative to the free nucleon is:

$$f_{\text{tick}} = \frac{BE}{m_p c^2} = \frac{8.8 \text{ MeV}}{938.272 \text{ MeV}} = 9.379 \times 10^{-3}. \tag{1}$$

This is numerically identical to the fractional mass deficit per nucleon in iron-56. Standard physics has held both values for a century in separate departments—nuclear physics

and general relativity—without placing them side by side. The medium framework makes the comparison direct: mass defect *is* tick-rate suppression.

Status: This is an observational identification grounded in two independently measured quantities. The numerical equivalence is exact; the claim that they represent the same physical phenomenon is the framework's central empirical assertion.

2.2 Contained and leaked fractions

Two distinct tick-rate suppression regimes coexist within the nucleus. The nuclear potential well depth ($\sim$ 50 MeV) produces an internal suppression of $\sim$ 5.33%, confined entirely within the nuclear boundary by the strong force. The binding energy (8.8 MeV) produces a suppression of 0.94% that crosses the nuclear boundary and propagates outward as the gravitational inflow field.

Table 1: Nuclear tick-rate suppression: contained vs. leaked.

Fraction	Energy	Suppression	Domain
Internal	$\sim$ 50 MeV	5.33%	Strong force
External	8.8 MeV	0.94%	Gravity

Internal: confined by strong force. External: propagates as gravity.

Strong force and gravity are therefore not separate interactions of different character. They are the contained and leaked fractions of one nuclear tick-rate suppression, partitioned at the nuclear boundary.

Status: Derived. The partition location is not assumed — it is where confinement ends. The strong force confines the internal fraction by definition; anything crossing the nuclear boundary propagates freely as the gravitational field. The numbers follow directly from measured binding energy and nuclear well depth. If the math works, that is where it happens.

2.3 Identification of the medium with the QCD vacuum

The derivation in Sec. 3.1 establishes that a mechanistic substrate must exhibit an inward flow at escape velocity to reproduce clock data. The quantum chromodynamic (QCD) vacuum is the candidate medium: a non-trivial, omnipresent field state responsible for $\sim$ 99% of hadronic mass [16].

The physical mechanism connecting matter to this flow is grounded in nuclear binding energy rather than QCD vacuum parameters directly. Nucleons release binding energy ($\sim$ 8.8 MeV per nucleon in iron-56) when bound into nuclei. This binding energy represents a suppression of the local tick rate—a fractional reduction $BE/m_p \approx 0.94\%$ that leaks outward through the nuclear boundary and propagates as the gravitational inflow field. The full derivation is given in Sec. 3.

This identification is proposed as a candidate consistent with known physics. The mathematical predictions in Secs. 3–9 follow from the flow field (Eq. 1) and tick-rate equation (Eq. 3) regardless of the medium's ultimate identity.

Status: Not derived. The QCD vacuum is the best available candidate — omnipresent, energetically significant, and structurally consistent — but no mechanism has been derived that connects QCD field equations to the gravitational flow field from first principles. The framework's predictions do not depend on this identification.

3 From Medium to Gravity

3.1 From clock data to flow field

Atomic clocks exhibit reproducible, precisely measured changes in frequency when placed at different gravitational potentials. If atomic tick rate depends on interaction with a back-

ground medium, then a spatial gradient in tick rate implies a corresponding gradient in the state of that medium. A minimal assumption is that the medium exhibits an effective flow relative to matter, and that the local tick rate depends on the relative velocity between matter and the medium.

The gravitational redshift formula provides the empirical constraint. If one asks what effective flow speed $v(r)$ would reproduce this tick-rate gradient through a velocity-dependent interaction, the solution is uniquely determined:

$$v(r) = \sqrt{\frac{2GM}{r}}, \tag{2}$$

the classical escape velocity. This result is not assumed; it follows directly from matching the observed tick-rate gradient to a velocity-dependent mechanism.

3.2 Effective velocity

An object near a mass has two sources of velocity relative to the medium: the inward flow of the medium itself (set by the local escape velocity) and the object's own motion. Both contribute to time dilation through the same mechanism. The total effective velocity combining these two contributions is:

$$v_{\text{eff}} = \sqrt{v_{\text{esc}}^2 + v_{\text{motion}}^2}. \tag{3}$$

This single equation unifies gravitational and kinematic time dilation: a clock sitting still near a massive body and a clock moving through empty space both slow down by the same mechanism—their effective speed through the medium.

3.3 Tick rate and refractive index

The tick rate $\hat{f}$ is the ratio of how fast a local clock runs to how fast it would run in empty flat space—a number between 0 and 1. It is precisely what GPS satellites correct for and what atomic clocks measure when moved to different altitudes. Higher effective velocity means lower tick rate, exactly as in special relativity:

$$\hat{f}(r) = \sqrt{1 - \frac{v_{\text{eff}}^2}{c^2}}. \tag{4}$$

The refractive index $n(r)$ is the inverse of the tick rate. It plays the same role here that the optical refractive index plays in optics: it determines how paths curve and how speeds appear to change. For a stationary observer in a gravitational field:

$$n(r) = \frac{1}{\hat{f}(r)} = \frac{1}{\sqrt{1 - r_s/r}}, \quad r_s = \frac{2GM}{c^2}. \tag{5}$$

One function of r controls all local physics: clock rates, ruler lengths, photon bending, and gravitational pull.

3.4 Gravitational acceleration

Objects fall because they follow gradients in the tick-rate field—moving toward regions where clocks run slower, just as light bends toward regions of higher refractive index. The local acceleration follows from the gradient of the tick-rate field:

$$g(r) = c^2 \hat{f} \frac{d\hat{f}}{dr}. \tag{6}$$

Substituting $\hat{f} = \sqrt{1 - r_s/r}$ gives

$$\frac{d\hat{f}}{dr} = \frac{1}{2}(1 - r_s/r)^{-1/2} \cdot \frac{r_s}{r^2}. \tag{7}$$

Therefore

$$g(r) = c^2 \cdot \sqrt{1 - r_s/r} \cdot \frac{1}{2\sqrt{1 - r_s/r}} \cdot \frac{r_s}{r^2} = \frac{c^2 r_s}{2r^2} = \frac{GM}{r^2}. \tag{8}$$

No force law is postulated. The Newtonian inverse-square law emerges automatically from the tick-rate gradient.

3.5 The balance equation

A stable circular orbit requires that the object's inertia exactly balances the tick-rate gradient pulling it inward. In the medium framework this condition sets the total effective velocity for a circular orbit—the escape velocity from the medium flow plus the orbital velocity combine to:

$$v_{\text{eff}}^{\text{circular}} = \sqrt{\frac{3GM}{r}}, \tag{9}$$

verified against solar system orbital data to 0.000000%.

3.6 The returnion: nuclear-to-gravitational crossover

The returnion r_{ret} is defined as the radial scale at which the gravitational potential energy between two nucleons equals the nuclear binding energy per nucleon. In the framework's natural language (κ, not G):

$$r_{\text{ret}} = \frac{\kappa c^2 m_p}{BE}. \tag{10}$$

Equivalently, substituting $G = \kappa c^2/m_p$: $r_{\text{ret}} = Gm_p^2/BE$.

Numerically:

$$r_{\text{ret}} = 1.324 \times 10^{-52} \text{ m}. \tag{11}$$

This is 53.3 times the proton's Schwarzschild radius $r_s(p) = 2Gm_p/c^2 = 2.484 \times 10^{-54}$ m.

The returnion marks the crossover scale: below r_{ret}, nuclear binding dominates; above r_{ret}, gravity dominates. In everyday terms: two nucleons closer together than r_{ret} are bound by the strong force; two nucleons farther apart than r_{ret} attract each other gravitationally. It is where nuclear geometry ends and spherical gravitational geometry begins—the boundary at which the $1/r$ nuclear field transitions to the $1/r^2$ gravitational field. The fact that this boundary exists and has a calculable value is a structural prediction of the framework.

3.7 Planetary mass from κ and N

The returnion enables a derivation of planetary mass from nuclear physics and a single kinematic observable (escape velocity). The only gravitational input is κ, measured from clocks.

The tick-rate suppression at a planet's surface, set by the planet's escape velocity, is:

$$f_B = \frac{v_{\text{esc}}^2}{2c^2}. \tag{12}$$

Each nucleon contributes one unit of tick suppression (f_{tick}) over one returnion length (r_{ret}). To produce the observed surface suppression f_B across the full planetary radius R, the required number of nucleons is:

$$N = \frac{f_B \cdot R}{f_{\text{tick}} \cdot r_{\text{ret}}}. \tag{13}$$

For Earth ($v_{\text{esc}} = 11{,}186$ m/s, $R = 6.371 \times 10^6$ m):

$$f_B = \frac{v_{\text{esc}}^2}{2c^2} = 6.961 \times 10^{-10} \tag{14}$$

$$N = \frac{f_B \cdot R}{f_{\text{tick}} \cdot r_{\text{ret}}} = 3.570 \times 10^{51} \tag{15}$$

$$M = Nm_p = 5.972 \times 10^{24} \text{ kg} \tag{16}$$

Translated into Newtonian language for comparison: $M_\oplus = 5.972 \times 10^{24}$ kg (measured). Agreement: 0.0004% when κ is computed from the laboratory value of G via $\kappa = Gm_p/c^2$. The derivation uses nuclear binding energy, proton energy, escape velocity, and planetary radius. No independent measurement of M is required.

3.8 Surface gravity from tick suppression

Surface gravitational acceleration follows directly:

$$g = \frac{f_B c^2}{R} = \frac{6.961 \times 10^{-10} \times c^2}{6.371 \times 10^6} = 9.820 \text{ m/s}^2. \tag{17}$$

Agreement with the measured value is 0.0001%.

3.9 Equivalence principle: structural proof

The equivalence principle follows structurally from the binding fraction. Object A, with mass m_A, has an interaction mass in the field of B equal to:

$$m_{\text{int}}(A) = m_A \times f_{\text{tick}}. \tag{18}$$

Its inertial mass—its resistance to acceleration—is also $m_A \times f_{\text{tick}}$, since both arise from the same binding energy fraction. The factor cancels:

$$a = \frac{m_{\text{int}}}{m_A}\, g_B = \frac{\cancel{f_{\text{tick}}} \cdot c^2 f_B/R}{\cancel{f_{\text{tick}}}} = \frac{GM_B}{R^2}. \tag{19}$$

The acceleration of A is independent of A's composition. Equivalence is not postulated—it is a structural consequence of interaction mass and inertial mass sharing the same origin. The small deviations predicted in Sec. 9.6 arise because f_{tick} is not perfectly uniform across nuclear species; the cancellation is exact only for identical compositions.

4 Formal Equivalence with General Relativity

4.1 The tick-rate metric

General relativity encodes gravity in the Schwarzschild metric—a set of coefficients describing how space and time are stretched near a mass. The key result here is that every coefficient in this metric is a function of the single refractive index $n(r)$ derived above. GR's full geometric machinery reduces to one number at each point in space:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \frac{dr^2}{1 - r_s/r} + r^2 d\Omega^2. \tag{20}$$

The time component $g_{00} = -c^2/n^2$ and radial component $g_{rr} = n^2$ are both determined by $n(r)$ alone. Particle paths (geodesics) become Fermat's principle: particles follow paths that extremize proper time through the refractive medium, exactly as light takes the shortest optical path.

4.2 Painlevé–Gullstrand connection

The medium flow at escape velocity corresponds exactly to the Painlevé–Gullstrand coordinate system [3, 4], in which the PG "rain" is not a coordinate artifact but a real velocity field in the medium.

4.3 The medium and Lorentz invariance

The medium introduces no preferred frame. Detection requires interaction; interaction requires relative motion; relative motion exists only where mass motivates flow or the detector accelerates. In flat spacetime both conditions are absent. All Lorentz-violation experiments in flat spacetime return null results to arbitrary precision—not because the medium is undetectable, but because there is nothing to detect.

4.4 Gravitational waves

Perturbations to the medium's velocity field propagate at c. GW170817 confirmed $c_{\text{grav}} = c_{\text{EM}}$ to $\sim 10^{-15}$ [11].

5 Experimental Tests

5.1 Prediction method

All predictions use κ and N as defined in Sec. 7. G never appears. The Sun's gravitational pull on Mercury is computed by counting the Sun's atoms and multiplying by κ.

5.2 Results

The framework has been tested against 15 independent measurements. All predictions derive from the flow field, effective velocity, tick-rate equation, and refractive index (Secs. 3.1–3.3) with no additional parameters. Selected results:

Table 2: Selected experimental tests.

Test	Predicted	Observed
GPS corrections	38.44 μs/d	38.4 μs/d
Solar deflection	1.750″	1.75″ ± 0.02″
Mercury prec.	42.988″/cy	43.0 ± 0.5″/cy
GW170817	$c_g = c_{\mathrm{EM}}$	Confirmed

Complete table of all 15 tests in Appendix C.

6 Nuclear and Atomic Structure

6.1 Baryon mass decomposition

Status: The fit is earned — four parameters, 12 baryons, $< 1.1\%$ RMS, and the proton mass emerges at 938.272 MeV without being used as input. The interpretation of constituent masses as medium processing costs is the framework's reframing of the existing DGG model [12]; this interpretation is not independently derived.

If nucleon mass is a cost of maintaining quark structure in the medium, then each quark species carries a characteristic maintenance cost ω_i, and pairs of quarks interact through a spin-dependent correction. The total baryon mass is the sum of quark costs plus spin interaction terms:

$$m_{\mathrm{baryon}} = \sum_i \omega_i + \sum_{\mathrm{pairs}} A \frac{\langle \vec{s}_i \cdot \vec{s}_j \rangle}{\omega_i \omega_j}. \tag{21}$$

Here $\langle \vec{s}_i \cdot \vec{s}_j \rangle = +1/4$ for parallel quark spins and $-1/2$ for antiparallel, and A is the spin-coupling strength. Four free parameters (ω_u, ω_d, ω_s, A) fit all 12 ground-state baryons to $< 1.1\%$ RMS (Appendix F).

6.2 Closed-circuit atomic mass fit

Status: Constrained fit, not a prediction. Eight parameters are fitted simultaneously to 43 measured atomic masses. The proton mass recovery at 938.272 MeV is the nontrivial result — it emerges from the fit structure without being supplied as an input. The 0.019% RMS error reflects the quality of the fit, not a derivation from first principles.

An atom with its electron shell forms what we term a "closed circuit": the nucleus provides the mass, the electrons provide the electromagnetic output channel, and the whole system is self-consistent. A bare proton or neutron outside a nucleus is an open circuit—no electron conduit, no electromagnetic closure. Modeling atomic mass as the sum of quark processing costs (feeding into nucleon masses), nuclear binding, and electron binding, eight parameters are fitted simultaneously to 43 atomic masses spanning hydrogen to uranium. No measured proton or neutron mass is used as input. Results: RMS error 0.019%, and the proton mass emerges from the fit at 938.272 MeV—matching the measured value to three decimal places.

6.3 Lepton masses and the Koide formula

Status: Observational pattern. The three-phase interpretation is suggested by the medium framework but not derived from it. A derivation would require a symmetry principle producing $\theta = 132.7°$ from first principles. This remains an open problem.

The three charged lepton masses satisfy the Koide relation [13] to 0.001%, with amplitude ratio $b = \sqrt{2}$ exact to 10^{-5}.

7 The Conversion Hierarchy

The contained and leaked fractions introduced in Sec. 2.2 define a quantitative hierarchy. The fraction of tick-rate suppression available for gravitational output scales with the local potential:

$$\frac{f_{\text{gravity}}}{f_{\text{total}}} = \frac{v_{\text{esc}}^2}{c^2} = \frac{r_s}{R}. \tag{22}$$

Gravity dominates at large scales because it is the only conversion product whose contributions superpose without cancellation. The hierarchy problem—why gravity is $\sim 10^{40}$ times weaker than the strong force—is resolved: they are the contained and leaked fractions of the same suppression, partitioned by the nuclear boundary.

8 The Per-Atom Tick Constant

8.1 Definition and measurement of κ

The framework's gravitational predictions use two quantities: κ, the per-atom tick constant, and N, the number of atoms. No other gravitational input is required.

κ is defined as the gravitational tick-rate contribution of a single atom—the total tick-rate difference across Earth's radius, divided by the number of atoms in Earth. Measured from surface clock data:

$$\kappa = \frac{\Delta \hat{f}_{\text{surface}} \times R_\oplus}{N_{\text{atoms}}} = 1.242 \times 10^{-54} \text{ m}. \tag{23}$$

This is measured once from any gravitational time-dilation experiment and never adjusted. Every gravitational prediction in this paper follows from κ and N alone:

$$g = N\kappa c^2/R^2 \tag{24}$$

$$v_{\text{esc}} = \sqrt{2N\kappa c^2/R} \tag{25}$$

$$\delta\phi = 6\pi N_\odot \kappa/[a(1-e^2)] \quad \text{(precession)} \tag{26}$$

8.2 Relationship to Newton's G

Newton's gravitational constant G is a translation constant between the atom-counting language of this framework and the kilogram language of Newtonian gravity:

$$G = \frac{\kappa c^2}{m_p}. \tag{27}$$

G has no independent physical status in this framework. It exists because Newton could not see atoms. He observed the net gravitational effect of $\sim 10^{50}$ atoms at once and had to package that collective behavior into a single constant. Every appearance of G in standard gravitational equations is a rewriting of $N\kappa c^2$:

$$GM = N\kappa c^2. \tag{28}$$

The force of gravity is real. The constant G is a unit conversion. This paper uses κ and N throughout; G appears only when translating results into Newtonian language for comparison with existing literature.

8.3 The causal chain

$$\begin{aligned} &\text{atomic structure} \to \tfrac{BE}{E_p} \to r_{\text{ret}} \\ &\quad \to \kappa \to v(r) \to \hat{f}(r) \to \text{gravity} \end{aligned} \tag{29}$$

Nuclear binding energy per nucleon sets the tick-rate suppression fraction BE/E_p. The returnion r_{ret} (Sec. 3) marks the crossover scale where this suppression transitions from the nuclear to the gravitational domain. κ is the per-atom gravitational tick constant measured from clocks. Clock data reads medium flow. Gravity reads tick rate. G is downstream of all of it—a translation constant, not a cause.

8.4 κ is derived, not assumed

The value of κ is not an unexplained empirical input like c or $\hbar$. It is determined by the framework's own chain of physical quantities,

and can be arrived at from two independent directions.

From atoms to gravity: The nuclear binding energy per nucleon (BE) sets the tick-rate suppression fraction $f_{\text{tick}} = BE/E_p$. The returnion $r_{\text{ret}} = \kappa c^2 E_p/(BE\, c^2)$ sets the crossover scale. Together, f_{tick} and r_{ret} determine how much tick-rate gradient each atom contributes to the gravitational field. κ is the output: it is what falls out when nuclear structure interacts with the medium.

From gravity to atoms: Clock data measures the tick-rate gradient at a known distance from a known body. Dividing by the number of atoms in that body gives κ. The same number, arrived at from the opposite direction.

The two paths converge because κ is not free. It is fixed by how much tick-rate suppression each nucleon produces, which is fixed by the binding energy, which is fixed by the nuclear physics. To say κ is "derived" means: given the nuclear binding energies (measured in spectrometers) and the medium's tick-rate response (measured by clocks), κ is determined. No additional assumption is needed. No free parameter is adjusted. The value 1.242×10^{-54} m is the number that nuclear structure and medium response jointly require.

The species-dependent decomposition (Sec. 8.5) makes this concrete: each species has its own $\kappa_A = \kappa \cdot E_A/(A\, E_p)$, and the single κ is the proton-energy calibration of this species-dependent quantity. The origin of κ is the origin of atomic binding energy—a nuclear physics question, not a gravitational one.

9 Discussion

9.1 Relationship to known physics

The framework's predictions are mathematically identical to GR in PG coordinates. The baryon model is identical to DGG [12]. What is new is the unified interpretation: quark processing costs, nuclear binding, electromagnetic structure, and gravitational flow as four manifestations of one medium.

9.2 What is earned and what is not

Earned: Formal equivalence with Schwarzschild geometry. All 15 experimental tests. Baryon masses. Atomic masses with emergent proton mass. Power partition and hierarchy resolution. Cosmological constant dissolution. Mass defect as tick-rate suppression (BE/E_p). Strong force and gravity as contained and leaked fractions. Returnion crossover scale. Planetary mass from κ and N (0.0004% with clock-derived κ). Equivalence principle from binding fraction cancellation. Species-dependent κ_A recovering g to six figures.

Not derived: Koide three-phase interpretation. Quark processing costs from Λ_{QCD}. Koide parameters (a, θ). Hawking radiation as failed processing loops.

9.3 The role of conceptual language

Every conceptual term maps to a quantitative equation: "tick suppression" maps to $f_{\text{tick}} = BE/m_p$; "leaked fraction" maps to the binding energy crossing the nuclear boundary; "conversion" maps to the partition fraction v_{esc}^2/c^2. Where a conceptual term appears without an adjacent equation, it is intended as interpretive motivation, not a quantitative claim.

9.4 Dynamical equations and the Lagrangian

The framework inherits the Einstein field equations through exact equivalence in the PG gauge. A medium-native Lagrangian can also be written directly, where $N_{cm}(r)$ is the number of atoms enclosed within radius r (the

"center of mass count"). Outside a body, N_{cm} is fixed at the total atom count. Inside a body, $N_{cm}(r)$ grows as r increases, accumulating layer by layer:

$$L = \frac{1}{2}\dot{r}^2 + \frac{N_{cm}(r)\,\kappa c^2}{r} \tag{30}$$

Applying the Euler–Lagrange equation produces the equation of motion:

$$\ddot{r} = -\frac{N_{cm}(r)\,\kappa c^2}{r^2} \tag{31}$$

G and M do not appear as separate quantities—they are absorbed into κ via $N\kappa c^2 = GM$, where κ is measured directly from clock data without independent knowledge of G or M. The equation of motion contains only the observable atom count and the clock-derived constant. This Lagrangian is self-similar across scales: the same equation governs a moon orbiting a planet and the tick-rate gradient from Earth's core to its crust, with only N_{cm} changing between the two cases.

9.5 Species-dependent κ and the composition of gravitational mass

Status: Unfinished. The decomposition below is exact within the framework and recovers standard gravitational predictions to full precision. The physical interpretation — that $G \times M$ is a shorthand for a species-weighted sum of atomic processing costs — is internally consistent but requires further development. This section is included to establish the structure; a complete treatment will follow.

The framework's single per-atom tick constant κ is an average. In reality, each atomic species has its own per-nucleon gravitational contribution. Since κ is the tick-rate contribution of a single proton-energy unit, the per-nucleon contribution for species with nucleon number A and measured atomic energy E_A (in MeV) is simply scaled by the energy ratio:

$$\kappa_A = \kappa\,\frac{E_A}{A\,E_p}, \tag{32}$$

where $E_p = 938.272$ MeV is the proton energy. No gravitational constant or mass appears — only a clock measurement (κ), an energy ratio from spectrometry (E_A/E_p), and a nucleon count (A). This is not a new gravitational effect — the composition dependence is already encoded in the total energy of any body. What the decomposition makes transparent is *why*: each species has a different processing cost in the medium, and the mass defect is the physical consequence of that cost difference.

The per-nucleon κ_A tracks the binding energy curve. Species near the iron peak, where binding energy per nucleon is highest, have the *lowest* κ_A because the most mass-energy has been released during nucleosynthesis and is no longer available as gravitational source. Light and heavy elements retain more energy per nucleon and therefore have higher κ_A:

Table 3: Species-dependent κ per nucleon from measured atomic energies (AME2020).

Species	E/A (MeV)	κ_A (10^{-54} m)	$\Delta\kappa/\kappa_{\mathrm{Fe}}$
H-1	938.78	1.2429	+8970 ppm
He-4	931.84	1.2336	−1527 ppm
O-16	930.94	1.2324	−566 ppm
Si-28	930.47	1.2318	+63 ppm
Fe-56	930.41	1.2317	0
Au-197	931.34	1.2329	+992 ppm
Pb-208	931.39	1.2330	+1051 ppm
U-238	931.69	1.2334	+1377 ppm

The total gravitational field of a composite body is the species-weighted sum:

$$g = \frac{c^2}{R^2}\sum_i N_i\,\kappa\,\frac{E_i}{E_p}, \tag{33}$$

where N_i is the number of atoms of species i and E_i is the measured atomic energy. Every quantity is either counted (N_i), measured from clocks (κ), measured from spectrometers (E_i/E_p), or measured with a ruler (R). No gravitational constant appears. No mass appears. In Newtonian translation, this is $g = GM/R^2$ with $M = \sum N_i m_i$.

For Earth's bulk composition (32.1% Fe, 30.1% O, 15.1% Si, 13.9% Mg, and trace

elements), the composition-weighted $\langle \kappa_A \rangle = 1.2320 \times 10^{-54}$ m. The species sum recovers $g = 9.8203$ m/s^2 to six significant figures — identical to the Newtonian result, as it must be, since the species sum is the atom-by-atom expression of the same physics.

The paper's single $\kappa = 1.242 \times 10^{-54}$ m, calibrated to the proton energy, differs from the composition-weighted value by 93 ppm — a small but real difference explained by the fact that Earth's average nucleon has less energy than a free proton due to binding.

The returnion inherits this averaging: it uses E_p and a single BE when the physical system has species-dependent values. For iron-peak compositions the average is excellent. The framework's natural expression is the species sum $\sum N_i \kappa (E_i/E_p)$, of which $N\kappa$ is the single-species approximation.

9.6 Equivalence principle as approximation: quantitative prediction

Section 3 shows that the equivalence principle is structurally guaranteed: the binding fraction BE/m_p cancels exactly between interaction mass and inertial mass, making acceleration independent of composition. The framework's first prediction that differs from GR arises from the fact that BE/m_p is not perfectly uniform across nuclear species — different nuclei have slightly different values. A raw compositional difference therefore exists between test masses of different composition.

Status: Interpreted estimate. The structural proof (Sec. 3) is derived. The three-layer suppression calculation below is physically motivated and internally consistent, but the layer values are estimated from the framework's parameters rather than derived from a Lagrangian. The resulting $\eta \approx 4.3 \times 10^{-18}$ should be understood as an order-of-magnitude prediction, not a precision calculation. It is consistent with all current experimental bounds and provides a falsifiable target for next-generation tests.

Three successive suppression factors reduce the raw compositional difference to far below current experimental sensitivity. Each layer represents a physical mechanism that dilutes the signal:

Layer 1—Quark cost difference:

$$S_1 = \frac{\omega_d - \omega_u}{m_p} \approx \frac{4.8}{938} \approx 5.1 \times 10^{-3} \tag{34}$$

Layer 2—Nuclear binding suppression:

$$S_2 = \frac{B/A}{m_p} \approx 8.5 \times 10^{-3} \tag{35}$$

Layer 3—Tick-rate quadratic suppression:

$$S_3 = \left(\frac{v_{\rm esc}}{c}\right)^2 \approx 1.25 \times 10^{-9} \tag{36}$$

(evaluated at MICROSCOPE orbital radius).

For ^{48}Ti vs. ^{195}Pt:

$$\Delta f_n = 0.600 - 0.542 = 0.058 \tag{37}$$

$$\eta_{\rm raw} = \Delta f_n \cdot \frac{m_n - m_p}{\bar{m}} \approx 8.0 \times 10^{-5} \tag{38}$$

$$\eta_{\rm predicted} = \eta_{\rm raw} \times S_1 \times S_2 \times S_3 \approx 4.3 \times 10^{-18} \tag{39}$$

The MICROSCOPE bound is $|\eta| < 1.0 \times 10^{-15}$ [21]—two orders above the prediction. Consistent with all data. The deviation scales as:

$$\eta(r) \approx \eta_{\rm intrinsic} \times \left(\frac{v_{\rm esc}(r)}{c}\right)^2 \tag{40}$$

Table 4: EP deviation vs. gravitational environment.

Environment	$(v_{\rm esc}/c)^2$	η
MICROSCOPE orbit	1.25×10^{-9}	4.3×10^{-18}
Earth surface	1.38×10^{-9}	4.8×10^{-18}
White dwarf	2×10^{-4}	1.7×10^{-12}
Neutron star	0.31	2.7×10^{-9}

9.7 The cosmological constant dissolved

The cosmological constant problem—the 10^{120} discrepancy between QFT vacuum energy and the observed Λ—is conventionally treated as the worst fine-tuning problem in physics. The medium framework identifies it as a category error.

QFT vacuum energy calculations sum the zero-point energies $\hbar\omega/2$ of all quantum field modes. This is a *kinetic* quantity: it measures the energy cost of matter fields existing and fluctuating in the vacuum. It is correctly calculated within its domain.

The observed cosmological constant Λ, inferred from the accelerating expansion of the universe, measures something entirely different: the medium's baseline *potential* tension—the restoring force of the QCD vacuum substrate resisting spatial expansion.

Kinetic processing cost and potential tension are not the same physical quantity. They are not subject to comparison. The 10^{120} discrepancy is not a near-miraculous cancellation requiring explanation—it is the ratio between two fundamentally different things that happen to have very different magnitudes.

Furthermore, recent DESI results suggesting that Λ is not constant [22] are consistent with this picture. A true constant of the medium—its baseline potential tension—would be fixed. But the kinematic measurement of expansion rate evolves as the medium's restoration dynamics respond to the universe's growth. The medium has tension; expansion stretches it; the tension resists. Slowing expansion is not dark energy weakening—it is the medium's restoration dynamics catching up with the expansion rate.

Λ was never a property of the vacuum energy. It was a snapshot of the medium's restoration dynamics at a particular moment in cosmic history, mistaken for a constant because observations spanned insufficient time.

9.8 Open problems

1. Derive quark processing costs ω_u, ω_d from $\Lambda_{\mathrm{QCD}}, \alpha_s$.
2. Derive the Koide parameters (a, θ) from the medium's electromagnetic channel.
3. Extend to cosmological scales with a quantitative $w(z)$ prediction for comparison with DESI data.

10 Structural and Proximity Regimes of Gravitational Tick Rate

10.1 Motivation

The framework implicitly assumes the proximity regime in its experimental tests: the observer is outside the mass source, and tick rate varies with distance from it. This section shows that a second, physically distinct regime exists inside any mass distribution, and that it dominates the proximity regime by nearly four orders of magnitude at Earth's surface.

10.2 Two regimes from one Lagrangian

The per-atom Lagrangian (Sec. 8.4) produces two physically distinct regimes depending on whether $N_{cm}(r)$ is fixed or varying:

Outside—proximity regime: The enclosed atom count N_{cm} is fixed at the total. Tick rate decreases smoothly as you approach the body, like light slowing near a lens. Potential scales as $1/r$.

Inside—structural regime: As you move outward through the interior, each new shell of atoms adds to the enclosed count $N_{cm}(r)$, increasing it layer by layer. Tick rate is not set by distance from a point source—it is accumulated, like pressure building as you dive deeper in water, but here the "pressure" is the count of atoms overhead. For uniform density,

$N_{cm} \propto r^3$ and acceleration grows linearly with depth.

At the surface: The two regimes connect. The full tick-rate deficit built up through the interior projects outward and becomes the proximity field experienced by external observers.

10.3 Self-similarity across scales

The Lagrangian governing a moon orbiting a planet is structurally identical to the Lagrangian governing the tick-rate gradient from Earth's core to its crust. One equation. All scales.

10.4 Numerical verification for Earth

Using the PREM density profile, $\hat{f}(r) = \sqrt{1 - 2N_{cm}(r)\kappa c^2/(rc^2)}$:

Table 5: Tick rate profile through Earth's interior.

Location	v^2/c^2	$\hat{f}(r)$
Center	8.1×10^{-17}	≈ 1.000
Inner core boundary	1.2×10^{-10}	0.9999999999
Core–mantle bdry	8.3×10^{-10}	0.9999999996
Surface	1.4×10^{-9}	0.9999999993
1 km above surface	1.4×10^{-9}	0.9999999993
GPS orbit	1.2×10^{-8}	0.9999999938

$$\Delta\hat{f}_{\text{structural}} = 7.0 \times 10^{-10} \tag{41}$$

$$\Delta\hat{f}_{\text{proximity}} = 1.0 \times 10^{-13} \tag{42}$$

$$\text{Ratio} = 6{,}895 \tag{43}$$

10.5 Interpretation

> *Outside a mass: tick rate is a proximity effect, varying with distance from the enclosed source. Inside a mass: tick rate is a structural effect, built up layer by layer from center to surface. What we measure as gravitational time dilation near Earth's surface is the structural gradient of 6,371 km of accumulated mass, observed from its outer edge.*

An atom does not feel its own mass. By the shell theorem, self-mass contributions to the flow gradient cancel exactly at the atom's own location. Tick rate is set entirely by external mass.

11 Conclusion

Gravity exists as a force. This paper provides a physical cause for it: a medium flows toward mass at escape velocity, and atomic processes experience velocity-dependent resistance in that flow. The result is time dilation, spatial contraction, and gravitational acceleration—reproduced through a single refractive index that is formally equivalent to the Schwarzschild metric. The QCD vacuum is identified as the candidate medium.

The framework's natural language is κ (the per-atom tick constant, measured from clocks) and N (the number of atoms). Newton's gravitational constant G is a translation constant between this language and the kilogram language of classical gravity. It is not fundamental. The force is real; the constant is a unit conversion.

Extended to sub-nuclear scales, the framework fits 43 atomic masses from 8 configuration parameters to 0.019% accuracy, recovering the proton mass to three decimal places as an unconstrained output.

Nuclear binding energy per nucleon is identified as the tick-rate suppression fraction BE/m_p—a numerical equivalence between two quantities that have been measured independently for a century but never placed in direct correspondence. Strong force and gravity emerge as the contained and leaked fractions of a single nuclear suppression, partitioned at the nuclear boundary. The returnion $r_{\text{ret}} = \kappa c^2 E_p/(BE\, c^2)$ marks the crossover scale. Planetary mass follows from κ, N, and

escape velocity alone, with Earth's mass recovered to 0.0004%. The equivalence principle is structurally guaranteed by the exact cancellation of the binding fraction in both interaction and inertial mass.

The hierarchy between forces is resolved as a partition of tick-rate suppression determined by the nuclear boundary.

The cosmological constant problem is dissolved: QFT vacuum energy and Λ measure kinetic and potential quantities respectively—they are not subject to comparison. Slowing cosmic expansion, indicated by recent DESI results, is the medium's tension-restoration dynamics responding to stretching, not dark energy weakening.

Inside any mass distribution, tick rate is a structural effect set by accumulated enclosed mass, dominating the external proximity effect by $\sim$ 6,900$\times$ at Earth's surface. The same Lagrangian governs this internal structure and external orbital dynamics. One equation. All scales.

The framework predicts composition-dependent EP deviations at $\eta \sim 4 \times 10^{-18}$ for Ti vs. Pt—two orders below current sensitivity, with the signal growing quadratically with field strength. Whether the medium is ontologically real or a powerful computational framework, the quantitative results stand on their own.

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A Atomic Processing Power

Table 6: Partition of atomic processing power.

Environment	$v_{\rm esc}/c$	Matter state
Isolated atom	$\sim 10^{-20}$	Normal atom
Earth	3.7×10^{-5}	All elements
Sun	2.1×10^{-3}	Plasma
White dwarf	~ 0.014	e^--degenerate
Neutron star	~ 0.56	Nuclei dissolved
BH horizon	1.00	No structure

B Cost Hierarchy

Table 7: Electromagnetic cost hierarchy.

Tier	Coupling	Example
Strong/color	$\alpha_s \sim 1$	Confinement
EM (existence)	$\alpha \sim 1/137$	Electron mass
EM (binding)	$\alpha^2 \sim 10$ eV	Atomic structure

C Complete Experimental Tests

D Mercury Precession Derivation

$$\delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} = \frac{6\pi N_\odot \kappa}{a(1-e^2)} = 42.988''/\text{century} \tag{44}$$

E Complete Atomic Mass Table

Eight parameters fitted simultaneously to 43 measured atomic masses (AME2020 [15]). No measured atomic or nucleon masses used as input. Parameters: $\omega_u = 344.1$ MeV, $\omega_d = 348.9$ MeV, $A_s = 15.9 \times 10^6$ MeV^3, $a_V = 13.88$ MeV, $a_S = 10.57$ MeV, $a_C = 0.278$ MeV, $a_a = 96.21$ MeV, $a_P = 9.35$ MeV. RMS error: 0.019% (200 ppm) across all 43 atoms.

F Baryon Mass Table

Four parameters ($\omega_u = 362.2$ MeV, $\omega_d = 366.2$ MeV, $\omega_s = 537.2$ MeV, $A = 26.2 \times 10^6$ MeV^3) fitted to 12 ground-state baryons. RMS error $< 1.1\%$.

G Helium Ground State

$$E_{\rm He} = -2 \times Z_{\rm eff}^2 \times 13.598\ \text{eV} = -77.5\ \text{eV} \tag{45}$$

vs. experimental -79.0 eV (1.96%).

Table 8: All 15 experimental tests. All predictions derive from the core equations with no free parameters beyond κ, c, and N.

Test	Predicted	Observed	Agr.
NIST clocks (33 cm)	3.60×10^{-17}	$(4.1 \pm 1.6) \times 10^{-17}$	0.31σ
GPS corrections	38.44 μs/d	38.4 μs/d	0.10%
Pound–Rebka	2.46×10^{-15}	$(2.57 \pm 0.26) \times 10^{-15}$	0.45σ
Solar deflection	$1.750''$	$1.75'' \pm 0.02''$	$< 0.01\sigma$
Shapiro delay	Model	Cassini	1.1%
Mercury precess.	$42.988''$/cy	$43.0 \pm 0.5''$/cy	0.02σ
Venus precess.	$8.627''$/cy	$8.6 \pm 4.8''$/cy	0.01σ
Earth precess.	$3.839''$/cy	$3.8 \pm 1.2''$/cy	0.03σ
Mars precess.	$1.351''$/cy	$1.35 \pm 0.1''$/cy	0.01σ
Balance eq.	$\sqrt{3GM/r}$	Exact	0.0%
NS tick rate	0.7900	GR: 0.7900	0.0%
Hulse–Taylor	Model	$\dot{P}_b$	0.83%
GW170817	$c_g = c_{\mathrm{EM}}$	Confirmed	10^{-15}
He ground state	-77.5 eV	-79.0 eV	1.96%
Li ionization	5.40 eV	5.39 eV	0.1%

Table 9: Complete closed-circuit atomic mass predictions. Predicted values in MeV; deviation in ppm relative to AME2020 measured masses.

Atom	Z, N	Pred. (MeV)	Meas. (MeV)	ppm	Atom	Z, N	Pred. (MeV)	Meas. (MeV)	ppm
H-1	1,0	938.78	938.783	+0	P-31	15,16	28850.1	28851.6	−50
H-2	1,1	1876.59	1875.61	+250	S-32	16,16	29777.3	29781.8	−152
He-3	2,1	2808.50	2808.39	+37	Cl-35	17,18	32569.8	32573.3	−108
He-4	2,2	3728.73	3727.38	+361	Ar-40	18,22	37227.7	37224.3	+82
Li-6	3,3	5599.26	5601.52	−403	K-39	19,20	36289.5	36294.5	−136
Li-7	3,4	6530.92	6533.83	−446	Ca-40	20,20	37217.3	37224.7	−200
Be-9	4,5	8391.15	8392.75	−190	Ti-48	22,26	44665.5	44660.5	+111
B-10	5,5	9320.64	9324.44	−407	Cr-52	24,28	48384.8	48370.0	+305
B-11	5,6	10251.3	10252.5	−120	Fe-56	26,30	52104.3	52103.1	+24
C-12	6,6	11175.6	11174.9	+66	Ni-58	27,31	53966.6	53966.1	+9
C-13	6,7	12111.4	12109.5	+158	Cu-63	29,34	58618.3	58618.5	−4
N-14	7,7	13041.2	13040.2	+77	Zn-64	30,34	59544.0	59549.2	−86
N-15	7,8	13971.4	13968.9	+176	Se-80	34,46	74459.8	74441.3	+248
O-16	8,8	14896.6	14895.1	+104	Kr-84	36,48	78178.3	78163.5	+194
O-17	8,9	15831.3	15830.5	+53	Ag-107	47,60	99571.8	99600.1	−284
O-18	8,10	16764.0	16762.0	+116	Sn-120	50,70	111701	111703	+12
F-19	9,10	17691.2	17692.3	−61	I-127	53,74	118218	118175	+360
Ne-20	10,10	18617.1	18617.7	−33	Xe-132	54,78	122887	122868	+159
Na-23	11,12	21410.9	21409.2	+81	Au-197	79,118	183446	183473	−149
Mg-24	12,12	22337.3	22341.9	−208	Pb-208	82,126	193715	193729	−72
Al-27	13,14	25130.5	25126.5	+160	U-238	92,146	221700	221743	−193
Si-28	14,14	26057.3	26053.2	+158	*(continued above)*				

RMS: 0.019% (200 ppm). Proton mass emergent at 938.272 MeV. No measured masses used as input.

Table 10: Baryon mass predictions. Four parameters, 12 baryons, RMS $< 1.1\%$.

Baryon	Quarks	Pred. (MeV)	Obs. (MeV)	Err.
p	uud	942.8	938.3	$+0.5\%$
n	udd	945.7	939.6	$+0.7\%$
Λ	uds	1109.4	1115.7	-0.6%
Σ^+	uus	1179.5	1189.4	-0.8%
Σ^0	uds	1181.8	1192.6	-0.9%
Σ^-	dds	1184.2	1197.4	-1.1%
Ξ^0	uss	1329.2	1314.9	$+1.1\%$
Ξ^-	dss	1331.5	1321.7	$+0.7\%$
Δ	uuu	1237.4	1232.0	$+0.4\%$
Σ^*	uds	1381.3	1383.7	-0.2%
Ξ^*	uss	1527.6	1531.8	-0.3%
Ω^-	sss	1672.5	1672.5	$+0.0\%$